Scheme and Answer Key -QP CODE: 25900558

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPHY200 – BASIC SEMICONDUCTOR PHYSICS

Part A

Answer any **10** questions, Each question carries **2** marks

1. Write the relation of transition capacitance with reverse bias voltage

$C_T = \frac{K}{(V_K + V_R)^n}$	1.5 Mark
V_K – *Knee voltage*, V_R – *Reverse voltage*, K – *Constant*	0.5 *Mark*

2. Write the values of barrier potential of Silicon PN Junction and Germanium PN Junction at room temperature.

Silicon (Si): 0.7V	1 Mark
Germanium (Ge): 0.3V	1 Mark

3. List any two applications of a Zener diode

Voltage Regulation	Any **TWO** applications – 2 Marks
Fixed reference voltage	
Clipper circuit	
Protection against excessive voltage	

4. What is dynamic resistance of a diode?

Dynamic resistance is the small change in voltage divided by the corresponding change in current at a given operating point on the diode's I–V curve OR Dynamic resistance is the sum of bulk and junction resistance	Statement (2 Marks) OR Formula (2 Marks) OR Representation in V-I plot(2 Marks)
$r_d = \frac{dV}{dI}$ or $r_d = r_B + r_j$	

5. Draw the V–I characteristics curve for a PN junction diode.

V-I curve with forward bias region **only**	1.5 Marks
V-I curve with forward and reverse bias regions	2 Marks

6. A full-wave rectifier gives a DC output of 50V and RMS value of ripple 21V.Find ripple factor.

$Ripple\ factor, \gamma = \frac{V_{rms}}{V_{dc}}$	1 Mark
$\gamma = \frac{21}{50} = 0.42$	1 Mark

7. Why is half-wave rectifier not preferred for power supply applications?

Low efficiency	1 Mark
High ripple factor	1 Mark

8. What is the difference between Peak Inverse Voltage of center-tapped and bridge rectifier.

PIV of Center-tapped rectifier =$2V_m$	1 Mark
PIV of Bridge rectifier =V_m	1 Mark

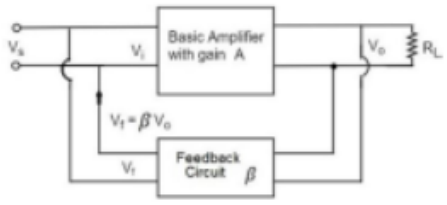
9. Why is it important to choose Q-point at middle of load line?

To allow maximum output swing without distortion	Any **TWO** points – 2 Marks
Keeps the transistor operating in active region	
Prevents cutoff or saturation of the transistor during signal variations	
Ensures faithful amplification	

10. Define α and β in transistor configuration.

$\alpha = \frac{I_c}{I_E}\ or \frac{\Delta I_c}{\Delta I_E}$	1 Mark
$\beta = \frac{I_c}{I_B}\ or \frac{\Delta I_c}{\Delta I_B}$	1 Mark

11. Draw the block diagram of a voltage-shunt feedback.

	1.5 Marks
Represent Input voltage , Amplifier, Feedback network in diagram	0.5 Mark

12. Define current gain of a CE amplifier.

The ratio of the collector current (I_C) to the base current (I_B).	1.5 Marks

$A_i = \beta = \frac{I_c}{I_B}$	0.5 Marks

13. Define bandwidth and gain-bandwidth product of an amplifier.

Bandwidth of an amplifier is the difference between 3 db frequencies. or $BW = f_2 - f_1$	1 Mark
Gain–Bandwidth Product : Product of the amplifier midband gain A and its bandwidth: GBP $= \mathrm{A} \times BW$.	1 Mark

14. Name the essential components of a feedback LC oscillator.

Tank circuit (A parallel connected inductor and capacitor)	1 Mark
Amplifier- a transistor that amplifies the signal	0.5 Mark
Feedback circuit- maintains continuous oscillations	0.5 Mark

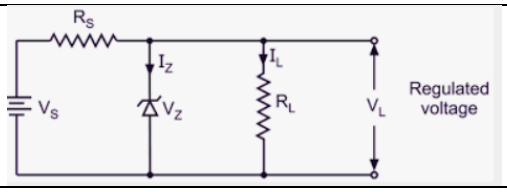
Part B

Answer any 6 questions, Each question carries 5 marks

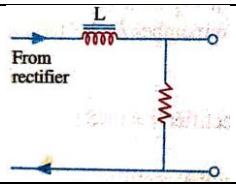
15. What happens when a PN junction is forward biased? Draw a forward biased PN junction?

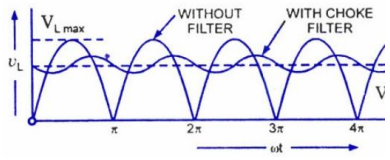
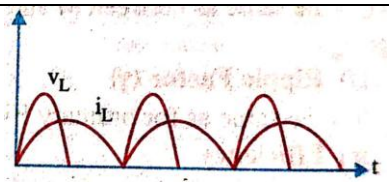
Forward bias reduces the width of the depletion region at the junction, lowering the potential barrier.	1.5 Marks
Once the applied voltage exceeds the barrier potential (~0.7 V for silicon), charge carriers can cross the junction, results in a continuous flow of current	1.5 Marks
Forward Biased PN Junction Diagram (Battery connected with positive to P-region and negative to N-region, showing reduced depletion region and current flow)	2 Marks

16. With neat circuit diagram explain the working of a Zener diode voltage regulator.

	2 Marks
When V_{in} > Zener breakdown voltage V_Z, the Zener diode maintains a constant voltage V_Z across R_L.	2 Marks
Any excess voltage is dropped across the series resistor R_S, and the Zener diode adjusts its current I_Z to keep the voltage stable	1 Mark

17. Draw the circuit diagram of an inductor filter and explain its working with waveform.

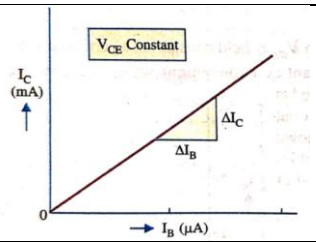
	2 Marks

An inductor allows low-frequency (DC) to pass easily and blocks high-frequency (AC) signals. So, it filters out the ripple from rectified output and gives smooth DC.	2 Marks
 or 	1 Mark

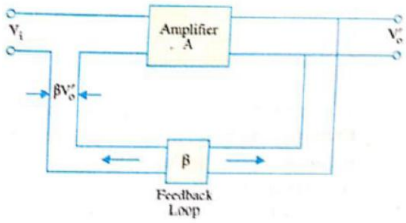
18. Explain the functions of emitter, base and collector regions of a transistor.

Block diagram/Symbol	1 Mark
Comparison of doping level and size of each region	2 Marks
Functions of each region	2 Marks

19. Draw transfer characteristics of a transistor in CE configuration. What is its importance?

	3 Marks
It shows how a small input current (I_B) controls a much larger output current (I_C).	1 Mark
Slope gives β value.	1 Mark

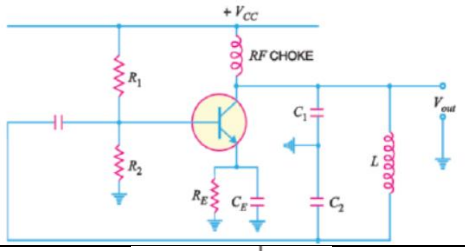
20. With the help of a diagram, explain the working principle of a feedback amplifier.

	2 Marks
$A\,(V_i + \beta V_0') = V_0'$ $A\,V_i = V_0' - A\beta V_0' = (1 - A\beta)V_0'$	2 Marks
Hence gain with feedback, $A' = \frac{V_0'}{V_i} = \frac{A}{1-A\beta}$ - for positive feedback And $A' = \frac{V_0'}{V_i} = \frac{A}{1+A\beta}$ - for negative feedback	1 Mark

21. In a CE transistor amplifier with voltage divider bias, show that the resistors R1 and R2 together with the emitter resistor could provide proper biasing and stability to the circuit.

The resistors R_1 and R_2 form a voltage divider that provides a fixed voltage to the base of the transistor irrespective of transistor β variations.	1 Mark
The voltage at the base V_2 is given by: $$V_2 = V_{CC} \times \frac{R_2}{R_1 + R_2}$$ This voltage is independent of the transistor's current gain.	2 Mark
The emitter resistor R_E provides negative feedback. The emitter current is $I_E = I_c = \frac{V_2 - V_{BE}}{R_E}$	1 Mark
If I_E tends to increase, the voltage drop across R_E increases, reducing the base-emitter voltage V_{BE}, which in turn reduces the emitter current back to normal.	1 Mark
Correct circuit diagram **only**	2 Marks

22. With necessary circuit diagram explain the working of Colpitts oscillator.

	2 Marks
$$f = \frac{1}{2\pi\sqrt{LC_T}}$$ $$C_T = \frac{C_1 C_2}{C_1 + C_2}$$	1 Mark
When powered on, small noise causes the tank circuit to start oscillating at its resonant frequency. The oscillating voltage across the capacitors is fed back to the transistor's input via the capacitor junction, providing the necessary positive feedback.	1 Mark
The transistor amplifies the signal, maintaining continuous oscillations. The total phase shift around the feedback loop is 360°	1 Mark